

# Datasheet-driven imperfection budgets for spectrally pure SPDC photon-pair sources

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Spectrally pure heralded single photons from spontaneous parametric down-conversion (SPDC) are usually engineered assuming an ideal pump laser. We translate three pump-laser datasheet quantities – center-frequency jitter, relative intensity noise (RIN), and longitudinal-mode structure – into a quantitative budget for heralded spectral purity. In a first-order group-velocity-mismatch model of a 10 mm PPKTP-like source, each imperfection enters as a Monte-Carlo ensemble of pure pump realizations whose averaged density matrix yields the heralded purity. The machinery is validated against an exact closed form for Gaussian phase matching, agreeing within error bars. Center jitter of  $27.7 \pm 0.3$ ,  $62.7 \pm 0.7$ , and  $91 \pm 1$  GHz rms consumes a 1%, 5%, and 10% relative purity budget; budgets scale as  $1/L$ , doubling for a re-matched 5 mm crystal and halving at 20 mm. One extra equal-power longitudinal mode at 0.25 THz spacing exceeds the 10% budget. RIN leaves the purity exactly invariant but inflates accidental double-pair emission as  $1+r^2$ . An explicit translation layer maps real datasheet lines – linewidth,  $d\lambda/dT$ , RIN in dB/Hz, cavity free spectral range – onto this budget for three commercial laser classes.

## I. INTRODUCTION

Spontaneous parametric down-conversion (SPDC) remains the workhorse process for generating photon pairs and, by detection of one photon of a pair, heralded single photons; it underpinned the first two-photon interference experiments [1] and the bright polarization-entangled sources that followed [2]. For modern applications the heralded photon must also be spectrally pure. When the joint spectral amplitude (JSA) of the pair carries signal-idler frequency correlations, the heralding measurement leaves its partner in a mixed spectral state, and photons from independent sources become partially distinguishable, degrading quantum interference [3]. The Schmidt decomposition makes this quantitative: the heralded purity is the inverse of the Schmidt number of the JSA [4]. Matching the group velocities of pump and down-converted fields can render the JSA approximately factorable at the source [5], a strategy demonstrated experimentally with heralded photons of approximately 95% purity produced without lossy spectral filtering [6].

A mature source-engineering literature has since pushed intrinsic purity toward unity, most prominently by apodizing the nonlinear response through domain engineering of the crystal [7, 8], and spectrally pure heralded sources are now a standard requirement in single-photon-source design [9]. Cross-comparisons of purity-characterization techniques on one and the same source have also shown that estimates based on the joint spectral intensity alone, blind to the joint spectral phase – pump chirp in particular – can substantially overestimate the purity a real source achieves [10].

In much of this work the pump laser is idealized as a noiseless, transform-limited, single-longitudinal-mode Gaussian, although the idealization is not universal. Experiments with a multimode cw diode pump have shown that the pump’s coherence structure transfers to the biphoton [11], and theory has traced how pump chirp maps onto the chirp and purity of the heralded sin-

gle photon [12]; the analogous transfer of partial pump coherence in the spatial (rather than spectral) domain has likewise been analyzed for the two-photon field [13]. These studies establish pump coherence as a phenomenon that propagates into the biphoton. What has been missing is the engineering translation: a mapping from the specifications printed on a commercial laser datasheet – center-frequency drift and jitter, relative intensity noise (RIN), and the longitudinal-mode structure of the laser cavity – to a quantitative purity budget with stated uncertainties. Table I positions the present work against these studies and their context: each addresses a single imperfection channel or an adjacent design aspect, but none supplies the datasheet-level translation, an uncertainty-quantified budget, or an external closed-form check. How many gigahertz of pump jitter cost one percent of heralded purity? Does RIN matter at all?

This paper provides that translation for a concrete, numerically transparent model. We study a collinear SPDC source described by first-order group-velocity-mismatch (GVM) phase matching, with parameters representative of a 10 mm PPKTP crystal pumped at 775 nm and emitting at 1550 nm, with a transform-limited Gaussian pump of 0.164 THz bandwidth (standard deviation of the amplitude envelope); the ideal heralded purity of this reference design is  $P = 0.810$ . Each pump imperfection is mapped to a Monte-Carlo ensemble of pure pump realizations, and the heralded photon is described by the ensemble-averaged density matrix, from which we compute the mixed-state purity directly. The Monte-Carlo machinery is validated externally against an exact closed-form purity for Gaussian phase matching, with which it agrees within error bars (Fig. 4). The headline of the resulting imperfection budget (Table II) is that center jitter of  $27.7 \pm 0.3$ ,  $62.7 \pm 0.7$ , and  $91 \pm 1$  GHz rms (mean  $\pm$  SEM over 16 Monte-Carlo seeds) consumes a 1%, 5%, and 10% relative purity budget, respectively – the 1% figure about one sixth of the 0.164 THz (164 GHz) pump bandwidth. More broadly, datasheet lines map onto two dis-

| Work                              | Imperfection or context studied                          | Method     | Spec translation | Budget $\pm$ uncert. | Closed-form check |
|-----------------------------------|----------------------------------------------------------|------------|------------------|----------------------|-------------------|
| Grice & Walmsley [3]              | bandwidth (transform-limited)                            | theory     | —                | —                    | —                 |
| Kwon <i>et al.</i> [11]           | multimode cw diode pump                                  | experiment | —                | —                    | —                 |
| Jha & Boyd [13]                   | spatial pump coherence                                   | theory     | —                | —                    | —                 |
| Sánchez-Lozano <i>et al.</i> [12] | pump chirp                                               | theory     | —                | —                    | —                 |
| Takeoka <i>et al.</i> [14]        | multi-pair statistics                                    | theory     | —                | —                    | —                 |
| Graffitti <i>et al.</i> [8]       | crystal side (apodization)                               | experiment | —                | —                    | —                 |
| This work                         | jitter, mode structure, RIN — as printed datasheet lines | numerics   | ✓                | ✓                    | ✓                 |

TABLE I. Qualitative positioning relative to prior studies. Earlier work established that individual pump imperfections – bandwidth, multimode structure, chirp, spatial coherence – propagate into the biphoton, and adjacent studies treated multi-pair statistics and crystal-side apodization; this work adds the engineering layer those studies leave open: a translation from printed datasheet lines to model parameters, a purity budget with stated Monte-Carlo uncertainties, and an external closed-form validation of the numerical pipeline.

tinct damage channels: spectral imperfections (jitter and longitudinal-mode structure) that degrade the heralded purity, and statistical imperfections (RIN) that leave the purity exactly invariant but degrade the pair statistics.

The scope is that of a numerical modeling study: first-order (linear) GVM phase matching, quasi-static ensemble noise, and heralded spectral purity (together with the relative double-pair boost) as the figures of merit. Absolute pair rate, heralding efficiency, and higher-order dispersion lie outside the model.

This paper makes four contributions.

1. The first quantitative pump-imperfection budget for heralded spectral purity with stated Monte-Carlo uncertainties (Table II, Sec. III).
2. An explicit datasheet-to-model translation layer, applied to three real commercial lasers (Table III, Sec. IV).
3. Closed-form external validation of the ensemble pipeline against the exact double-Gaussian purity of Eqs. (7) and (8) (Sec. II; Fig. 4).
4. A demonstrated  $1/L$  scaling law for the jitter tolerances (Fig. 5).

## II. MODEL AND METHODS

### A. Joint spectral amplitude

We consider collinear, single-spatial-mode SPDC and describe the emitted photon pair by its joint spectral amplitude (JSA). Writing  $\nu_s$  and  $\nu_i$  for the angular-frequency detunings of signal and idler from their central design frequencies, the JSA factorizes into an energy-conservation term set by the pump and a momentum-conservation term set by the crystal [3],

$$f(\nu_s, \nu_i) = \alpha(\nu_s + \nu_i) \phi(\nu_s, \nu_i). \quad (1)$$

For a transform-limited Gaussian pump the envelope is  $\alpha(\nu) = \exp[-\nu^2/(2\sigma^2)]$ , a band along the antidiagonal  $\nu_s + \nu_i = 0$ ;  $\sigma$  is the standard deviation of the amplitude envelope in angular frequency. Bandwidths are quoted throughout as the corresponding ordinary-frequency values; the reference pump has  $\sigma/2\pi = 0.164$  THz. For the phase matching we use the standard first-order group-velocity-mismatch (GVM) expansion of the wave-vector mismatch in a crystal of length  $L$ ,

$$\phi(\nu_s, \nu_i) = \text{sinc}\left(\frac{\Delta k L}{2}\right), \quad \Delta k = \kappa_s \nu_s + \kappa_i \nu_i, \quad (2)$$

with  $\text{sinc}(x) = \sin(x)/x$  and  $\kappa_j = 1/v_g^{\text{pump}} - 1/v_g^{(j)}$  the group-velocity mismatch between the pump and photon  $j$ . The reference crystal has  $L = 10$  mm,  $\kappa_s = +0.207$  ns/m, and  $\kappa_i = -0.318$  ns/m, representative of PPKTP pumped at 775 nm and producing degenerate pairs at 1550 nm [3, 5, 8]. The ratio  $\kappa_s/\kappa_i$  sets the tilt of the sinc ridge in the  $(\nu_s, \nu_i)$  plane, and the interplay of this tilt with the pump band determines how separable the JSA can be (Fig. 1). We deliberately retain only this first-order model: no Sellmeier dispersion, no higher-order terms, and no apodization of the nonlinearity [7, 8], so that the imperfection budgets below isolate pump effects at a fixed, transparent phase-matching geometry.

### B. Schmidt decomposition

The JSA is evaluated on a square grid of  $n$  points per axis over a window of half-width 4–8 THz (chosen per study so that the intensity decays to zero well inside the boundary) and normalized so that  $\sum |f|^2 = 1$ . The Schmidt decomposition of the discretized JSA is its singular-value decomposition [4]: with singular values  $s_k$ , the Schmidt coefficients are  $\lambda_k = s_k^2 / \sum_j s_j^2$ , and the heralded-photon spectral purity and Schmidt number are

$$P = \sum_k \lambda_k^2, \quad K = 1/P. \quad (3)$$

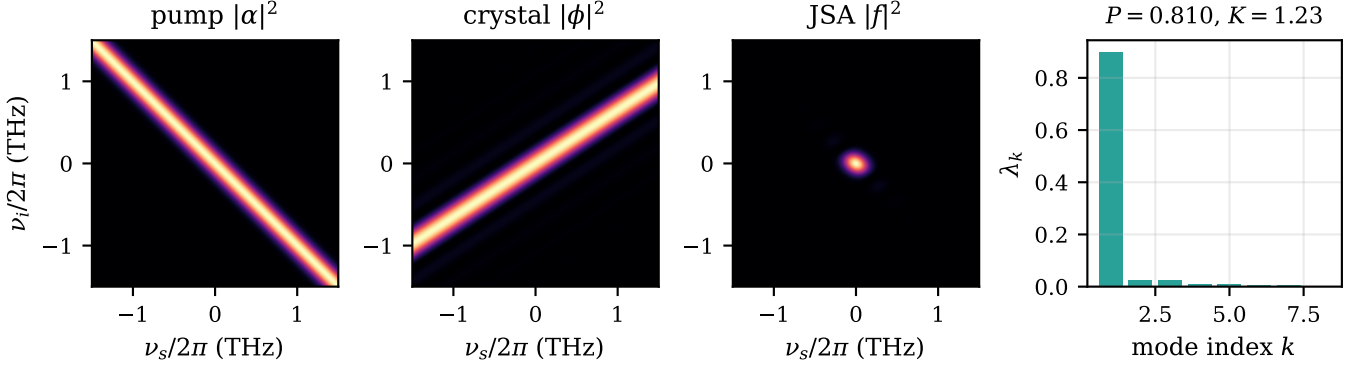


FIG. 1. Anatomy of the JSA at the reference point ( $L = 10$  mm,  $\sigma/2\pi = 0.164$  THz); axes are detunings  $\nu/2\pi$  in THz. Left to right: pump intensity  $|\alpha|^2$ , a band along the antidiagonal  $\nu_s + \nu_i = 0$  (energy conservation); phase-matching intensity  $|\phi|^2$ , a sinc ridge with side lobes whose tilt is set by  $\kappa_s/\kappa_i$  (momentum conservation); their product, the joint spectral intensity  $|f|^2$ , a compact and nearly isotropic central lobe with faint sinc side lobes; and the Schmidt spectrum  $\lambda_k$  of the discretized amplitude, dominated by a single coefficient and giving  $P = 0.810$ ,  $K = 1.23$ .

$P = 1$  ( $K = 1$ ) corresponds to a fully separable JSA. At the reference point  $P = 0.8100$  ( $K = 1.2345$ ).

### C. Imperfect pump as a mixed-state ensemble

An imperfect pump is a classical ensemble of pure pump realizations, so the heralded signal photon is described by a density matrix rather than a single Schmidt spectrum. We Monte-Carlo sample  $N$  pure realizations per sweep point ( $N = 120$  in the standard sweeps,  $N = 240$  in the fine-resolution jitter study, and  $N = 1000$  in the RIN sweep, whose per-realization cost is lower), compute the discretized JSA matrix  $A_k$  for each, and accumulate

$$\rho \propto \sum_k w_k A_k A_k^\dagger, \quad (4)$$

from which the heralded purity follows in a normalization-independent form,

$$P = \frac{\text{Tr}(\rho^2)}{[\text{Tr}(\rho)]^2}. \quad (5)$$

Each imperfection is parameterized directly by laser datasheet statistics and enters only the pump. Per realization we sample: (i) *center jitter/drift* — a Gaussian shift of the pump center frequency with the quoted rms value, treated as a quasi-static shot-to-shot ensemble.

This quasi-static treatment has a definite regime of validity. The pump is *pulsed*:  $\sigma/2\pi = 0.164$  THz corresponds to picosecond-class pulses, and RIN in this model is pulse-energy noise. The ensemble picture holds for noise processes slower than the pump coherence time,  $\tau_{\text{coh}} \sim 1/\sigma \approx 1$  ps, so that each pair sees a static pump center, and faster than the integration time, so that the ensemble is fully sampled. For an 80 MHz-class source integrated over seconds this window spans noise frequencies from  $\sim$ Hz to tens of GHz and covers the dominant

laboratory processes — thermal drift, acoustic and mount vibrations, and drive-current noise. Sub-picosecond noise components instead act coherently on each pure JSA and lie outside the model.

(ii) *Longitudinal multimode structure* — one comb line drawn from a set of  $M$  discrete lines centered symmetrically on the pump, at offsets  $(m - (M - 1)/2)$  times the mode spacing for  $m = 0, \dots, M - 1$  (half-integer multiples of the spacing when  $M$  is even), with normalized line weights (uniform unless stated); the total pump shift is the sum of the comb-line offset and the jitter draw. Drawing a single line per realization asserts that the longitudinal modes are mutually *incoherent* — the correct description of a free-running multimode diode, whose modes carry random relative phases and exhibit mode-partition noise. It is not the correct description of a phase-coherent mode-locked comb: there the JSA is a coherent sum over the lines — a pure state, not a mixture — and the comb teeth, spaced by an 80 MHz-class repetition rate, are in any case far too fine to be resolved by the JSA.

(iii) *Relative intensity noise* (RIN) — a realization weight  $w = \max[0, 1 + \mathcal{N}(0, r^2)]$ , where  $\mathcal{N}(0, r^2)$  is a zero-mean Gaussian with standard deviation  $r$ , the fractional rms intensity noise, each realization's JSA being individually normalized.

Every Monte-Carlo sweep behind the figures and the budgets of Table II is run with  $S = 16$  independent seeds (the RIN sweep, whose heralded purity is exactly independent of the sampled weights — see the validation suite below — uses  $S = 8$ ). Auxiliary convergence checks — the single-seed grid-refinement replicas described below and the eight-seed  $N$ -doubling check of the jitter budgets — use fewer seeds. Each (seed, point) pair draws its  $N$  realizations from an independent random stream seeded by the pair itself; seeding streams by seed plus point index would instead make replicas overlap (seed  $s + 1$  at sweep point  $i$  reproducing seed  $s$  at point  $i + 1$ ) and thereby cor-

relate nominally independent replicas. Reported curves are means over the  $S$  per-seed curves, and the imperfection budgets of Table II are quoted as the mean  $\pm$  standard error of the mean (SEM) over the per-seed threshold crossings.

#### D. Closed-form benchmark: the double-Gaussian model

The Monte-Carlo machinery can be checked against mathematics rather than against itself. Replacing the sinc in Eq. (2) by a Gaussian phase-matching function [3, 8],

$$\phi_G(\nu_s, \nu_i) = \exp\left[-\gamma \left(\frac{\Delta k L}{2}\right)^2\right], \quad \gamma = 0.193, \quad (6)$$

makes the JSA a two-dimensional Gaussian and every quantity of interest analytic. The value of  $\gamma$  is fixed by matching the *amplitude* half-maximum of the sinc: both functions equal  $1/2$  at  $x = 1.8955$ . Note that because the crystal amplitude is  $\exp(-\gamma x^2)$ , the curvature entering the quadratic form below is  $2\gamma$ . With  $a_j = \kappa_j L/2$ , the JSA is  $f_G \propto \exp\{-\frac{1}{2}[A\nu_s^2 + 2C\nu_s\nu_i + B\nu_i^2]\}$  with

$$A = \frac{1}{\sigma^2} + 2\gamma a_s^2, \quad B = \frac{1}{\sigma^2} + 2\gamma a_i^2, \quad C = \frac{1}{\sigma^2} + 2\gamma a_s a_i,$$

and tracing out one photon gives the pure-state purity in closed form,

$$P_0 = \sqrt{1 - \frac{C^2}{AB}}. \quad (7)$$

For Gaussian center jitter of rms  $J$  (ordinary Hz) the mixed purity of Eq. (5) also closes:

$$P(J) = \frac{P_0}{\sqrt{1 + 4c(2\pi J)^2}}, \quad c = \frac{1}{2\sigma^2 + 1/(\gamma a_s^2)}. \quad (8)$$

Three points in the derivation are easy to get wrong. First, the cross term required for  $\text{Tr}(\rho^2)$  is  $\text{Tr}(AA^\dagger A' A'^\dagger) = \|A^\dagger A'\|_F^2$ , not the two-photon overlap  $|\langle f|f'\rangle|^2$ , which is the purity of the mixed *two-photon* state — a different quantity. Second, mixed reduced states are signal/idler *asymmetric*: the constant  $c$  carries the GVM constant of the *traced-out* photon, and since our  $\rho$  in Eq. (4) keeps the idler,  $c$  contains  $a_s$ . Third, a pump-center shift is an exact rigid translation of the JSA, so per-realization normalization is exact and the pair rate is independent of jitter in this model. At the reference point Eq. (7) gives  $P_0 = 0.9818$  — intrinsically more separable than the sinc model's  $P = 0.8100$ , whose side lobes carry entanglement — and exact separability ( $C = 0$ ,  $P_0 = 1$ ) occurs at a 0.1997 THz pump bandwidth, possible because  $\kappa_s \kappa_i < 0$ . Section III uses Eqs. (7) and (8) as an external validation of the full Monte-Carlo pipeline (Fig. 4).

#### E. From datasheet units to model parameters

Manufacturers publish spectral widths in MHz or nm, stabilities in pm or pm/K, RIN as a dB/Hz spectral density, and cavity geometry in mm — none of which the model accepts directly. The translation layer is deliberately explicit; each conversion carries one stated physical assumption. A Gaussian intensity-spectrum FWHM  $\Delta\nu_{\text{FWHM}}$  maps to the model bandwidth (the standard deviation of the amplitude envelope) as

$$\frac{\sigma}{2\pi} = \frac{\Delta\nu_{\text{FWHM}}}{2\sqrt{\ln 2}}. \quad (9)$$

Wavelength excursions convert at first order via

$$\delta\nu = \frac{c \delta\lambda}{\lambda^2}, \quad (10)$$

i.e. 499.1 GHz per nm at  $\lambda = 775$  nm. A bounded drift specification (“ $\pm a$  over  $T$  hours”, or a pm/K coefficient times a temperature stability) defines a peak-to-peak excursion band  $\text{pp} = 2a$  rather than a Gaussian; taking the center uniformly distributed over the band gives

$$\text{rms} = \frac{\text{pp}}{2\sqrt{3}}. \quad (11)$$

A RIN spectral density, assumed white over the collection bandwidth  $B$  (the inverse of the detection gate or pulse period), integrates to the fractional rms

$$r = \left[10^{\text{RIN}/10} B\right]^{1/2}, \quad (12)$$

and a standing-wave laser cavity of length  $L_{\text{cav}}$  and refractive index  $n$  has longitudinal-mode spacing

$$\text{FSR} = \frac{c}{2nL_{\text{cav}}}, \quad (13)$$

which maps onto the comb spacing of the multimode model.

One published number deserves a caveat rather than a conversion: a Lorentzian linewidth arises from fast phase diffusion and has no finite second moment, so no rms exists and the quasi-static ensemble does not describe it — phase noise that decorrelates within a pulse does not rigidly displace the pump band. We therefore use the Lorentzian HWHM only as a deliberately conservative stand-in bound on the center jitter; for the MHz-class linewidths of DFB diodes this bound sits orders of magnitude below the GHz-scale budgets of Table II and is negligible. Slow frequency-drift specifications ( $\pm \text{pm}$  over hours, pm/K times a temperature stability) are quasi-static and enter through Eqs. (10) and (11) instead. These conversions are applied to real products in Table III.



## F. Numerical robustness and validation

Ideal-pump sweeps use  $n = 512$ ; Monte-Carlo studies use  $n = 256$ , which is converged to better than  $10^{-4}$  in  $P$  against  $n = 768$  at the hardest points of the parameter sweeps (0.05 THz bandwidth over a 6 THz half-window; 2 mm crystal over an 8 THz half-window). The RIN sweep uses a coarser  $n = 64$  grid: there every realization shares the identical JSA and only the scalar weights vary, so the reported purity invariance is exact at any grid size. Single-seed replicas of the 10 mm jitter sweep at  $n = 128$  and  $n = 192$  shift the mixed purity by only  $\sim 2 \times 10^{-5}$ , far below the seed-to-seed Monte-Carlo spread: sampling noise, not discretization, dominates the error budget. The code ships with a validation suite of fourteen physics tests, including exact collapse of the zero-imperfection ensemble to the pure-state SVD purity ( $P = 0.8100$ ), exact RIN invariance of the heralded purity (difference below  $10^{-9}$ ), reduction of a comb with zero side-line weight to the single-mode result, monotonic degradation of purity with jitter, and closed-form benchmarks against the double-Gaussian model: the SVD purity matches Eq. (7) to better than  $10^{-4}$ , and the full Monte-Carlo pipeline reproduces Eq. (8) including its finite- $N$  estimator bias.

All simulations use the open-source `qsource-sim` package (Python, MIT license), available at <https://github.com/ABD252/qsource-sim>; a Zenodo DOI will accompany submission. Every figure and table in this paper regenerates from a declarative YAML configuration file, one per study.

## III. RESULTS

### A. Ideal-source baseline

We first establish the ideal-pump baseline that all imperfection budgets refer to. Figure 2(a) shows the heralded purity of Eq. (5) as a function of the pump bandwidth for the 10 mm reference crystal. The curve has the broad interior optimum expected when the pump-envelope and phase-matching correlations of Eq. (1) are balanced [3, 4, 6]: the grid-limited peak is  $P_{\max} = 0.8106$  at 0.178 THz on the sweep's grid ( $n = 512$ , 6 THz frequency half-window), the top is flat, and the canonical reference point  $\sigma = 0.164$  THz gives  $P = 0.8100$  on the reference grid ( $n = 256$ , 4 THz half-window) used for all Monte-Carlo studies below. The 1% tolerance window spans 0.159–0.202 THz, so the bandwidth itself is a forgiving design parameter. The fall-off is nearly log-symmetric about the optimum: equal factors from the peak give nearly equal purities (e.g.  $P = 0.663$  and  $0.665$  a factor of 1.9 below and above it), and the endpoint values— $P = 0.41$  at 0.05 THz against 0.24 at 1.2 THz—differ only because the sweep extends further on the broadband side (factors of 3.6 and 6.7 from the peak, respectively). The residual gap to  $P = 1$  is the familiar penalty of an unapodized sinc phase-matching function

[7, 8]. Figure 2(b) shows the complementary sweep: at fixed  $\sigma = 0.164$  THz the crystal length has its optimum at 10–12 mm ( $P = 0.806$  at 10 mm, 0.805 at 12 mm), falling to 0.286 at 2 mm and 0.51 at 30 mm. This is the same matching condition seen from the crystal side. The small offset between the 0.806 quoted here and the 0.8100 reference value above is a frequency-window truncation effect, not discretization: the slowly decaying sinc side lobes make the computed purity drift with the window half-width ( $P = 0.8100$ , 0.8074, and 0.8061 at the same operating point for the 4, 6, and 8 THz half-windows of the reference grid, the bandwidth sweep, and the length sweep, respectively), so ideal purities are comparable only within a fixed window and are window-independent only at the level  $P \approx 0.81$ . As a narrowband anchor, a 5 GHz pump yields  $P = 0.044$  and  $K = 22.6$  (Eq. (3)) once the grid resolves the few-GHz pump ridge (which is narrower than the pixel of the sweep grids), recovering the strongly anti-correlated quasi-CW limit.

### B. Spectral drift and jitter

Figure 3 shows the heralded purity of the Monte-Carlo mixed state (Eq. (4)) as the pump center frequency jitters with increasing rms amplitude, for three length-matched sources. At zero jitter each ensemble reproduces its ideal value exactly ( $P(0) = 0.8100$  for the 10 mm reference), and the seed-mean degradation is monotonic across the sweep. The shaded bands are the  $\pm 1\sigma$  spread over  $S = 16$  independent seeds ( $N = 120$  realizations per seed and sweep point); this seed-to-seed scatter dominates the negligible discretization error ( $\sim 2 \times 10^{-5}$  under grid refinement at fixed window). Quoting each budget as the mean  $\pm$  standard error of the per-seed threshold crossings, the 10 mm reference crystal tolerates  $27.7 \pm 0.3$ ,  $62.7 \pm 0.7$ , and  $91 \pm 1$  GHz rms of center jitter for a 1/5/10% relative purity drop (in wavelength terms, 27.7 GHz at 775 nm is 55.5 pm). The 1% cell comes from a dedicated fine sweep (5 GHz steps around the crossing,  $N = 240$ ); an  $N$ -convergence check on the 10 mm sweep finds  $|b(N=240) - b(N=120)| < 2 \text{ SEM}$  at every drop level, so the budgets are unchanged within error bars when  $N$  doubles. Re-matching the source with a 5 mm crystal and a 0.328 THz pump (ideal  $P = 0.8183$ ) roughly doubles every budget, to  $54.1 \pm 0.9$ ,  $125 \pm 1$ , and  $181 \pm 3$  GHz, while a 20 mm crystal matched to a 0.082 THz pump (ideal  $P = 0.8061$ ) roughly halves them, to  $13.5 \pm 0.2$ ,  $31.3 \pm 0.4$ , and  $45.3 \pm 0.6$  GHz: the phase-matching bandwidth scales as  $1/L$  (Eq. (2)), so a shorter crystal simply presents a wider acceptance to a wandering pump, at the cost of a lower pair rate that our model does not capture. We quantify this scaling below.

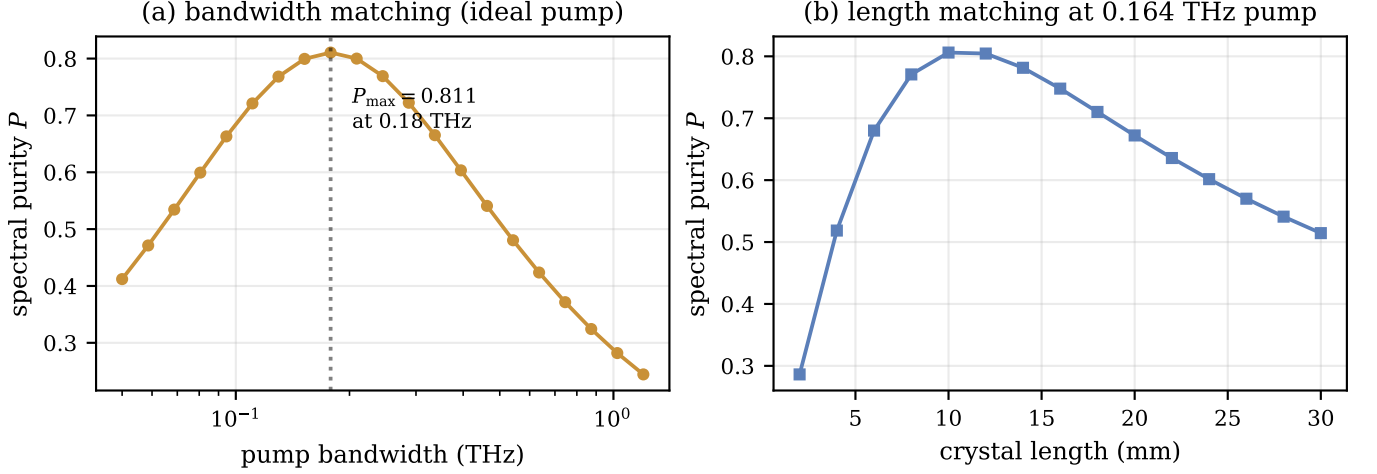


FIG. 2. Ideal-source design space. (a) Heralded purity versus pump bandwidth (log axis) for the 10 mm crystal; the dotted line marks the grid-limited peak  $P_{\max} = 0.8106$  at 0.178 THz on a flat-topped optimum. (b) Purity versus crystal length (2–30 mm) at the fixed reference pump bandwidth of 0.164 THz, with the optimum at 10–12 mm.

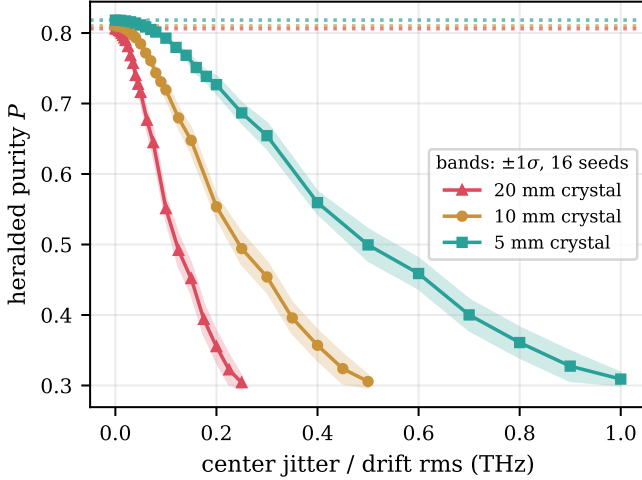


FIG. 3. Heralded purity versus pump center jitter/drift (rms, treated as a quasi-static ensemble) for three length-matched sources: a 20 mm crystal with a 0.082 THz pump (triangles), the 10 mm reference source (circles), and a 5 mm crystal with the pump re-matched to 0.328 THz (squares). Shaded bands are the  $\pm 1\sigma$  spread over 16 Monte-Carlo seeds; dotted lines mark the respective ideal purities.

### C. External validation against a closed form

The Monte-Carlo pipeline can be benchmarked against an exact result. For Gaussian phase matching the JSA is a two-dimensional Gaussian and remains one under any rigid center shift, so the purity of the jitter ensemble takes the closed form  $P(J) = P_0 [1 + 4c(2\pi J)^2]^{-1/2}$  of Eq. (8), with  $J$  the rms center jitter and  $P_0 = 0.9818$  at the reference operating point. Figure 4(a) overlays the full Monte-Carlo sweep on this prediction: the  $\pm 1\sigma$  band of 16 seeds lies on the curve across the en-

tire sweep, and the budgets extracted from the Monte-Carlo data by the same threshold machinery used everywhere else ( $30 \pm 0.3$ ,  $68.5 \pm 0.8$ , and  $102 \pm 1$  GHz for 1/5/10%) match the closed-form predictions (29.9, 69.0, and 101.7 GHz) within error bars. This validates the estimator end to end—sampling, mixing, and threshold extraction—against a result derived with no Monte-Carlo input. For the sinc model no closed form exists, but its normalized decay  $P(J)/P(0)$  tracks the analytic shape  $g(J) = [1 + 4c(2\pi J)^2]^{-1/2}$  with a maximum deviation of 0.041 over the dense sweep (Fig. 4(b)). We emphasize what this comparison does and does not show: the Gaussian model’s ideal purity ( $P_0 = 0.9818$ ) differs strongly from the sinc value (0.8100) because the unapodized sinc side lobes carry entanglement [7, 8], so panel (b) compares decay *shapes* only, never absolute purities. Consistently, the Gaussian-PM jitter budgets are 8–13% larger than the sinc budgets (about 8% at the 1% level, e.g.  $30 \pm 0.3$  against  $27.7 \pm 0.3$  GHz, rising to 13% at the 10% level): the side lobes make the unapodized source slightly more jitter-sensitive. The ensemble method itself transfers unchanged to engineered (apodized) crystals [7, 8], whose phase matching is close to this Gaussian benchmark.

### D. Scaling with crystal length

The three length-matched jitter sweeps collapse onto a simple design law. Figure 5 plots the jitter budgets against the inverse crystal length: the 5% budgets of 31.3, 62.7, and 125 GHz at  $L = 20, 10$ , and 5 mm stand in the ratios 1 : 2.00 : 4.00, and through-origin linear fits describe all three drop levels within twice the seed-to-seed standard errors. This is a demonstrated  $1/L$  law, consistent with the  $1/L$  scaling of the phase-matching

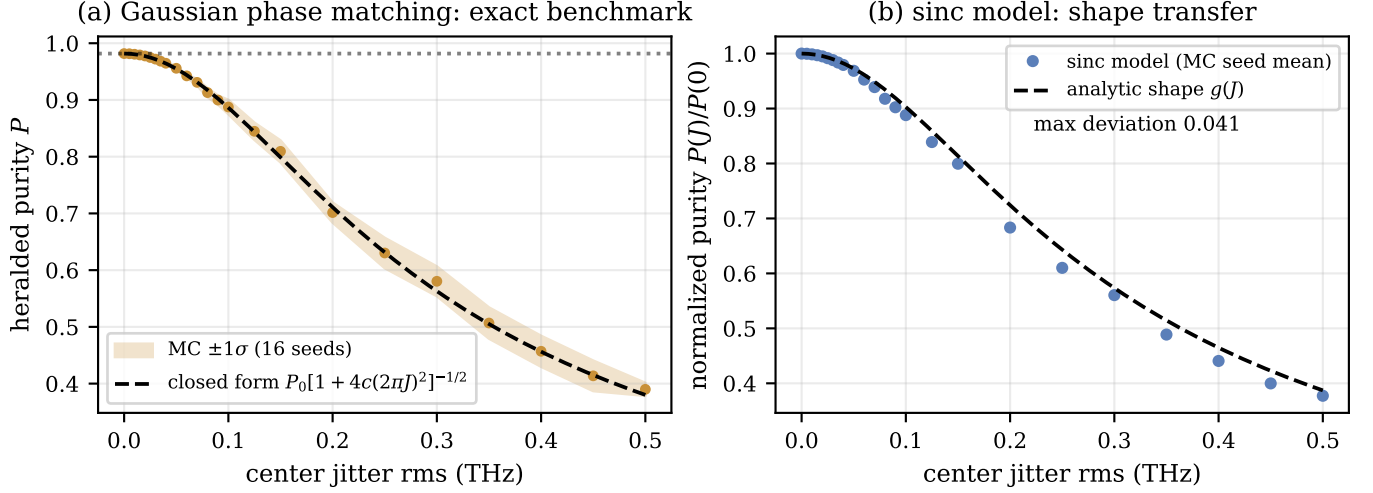


FIG. 4. External validation against a closed form. (a) Gaussian phase matching: heralded purity versus center jitter from the full Monte-Carlo pipeline ( $\pm 1\sigma$  band over 16 seeds) against the closed form  $P_0[1 + 4c(2\pi J)^2]^{-1/2}$  of Eq. (8) (dashed); the dotted line marks the ideal purity  $P_0 = 0.9818$ . (b) Shape transfer to the sinc model: normalized decay  $P(J)/P(0)$  (seed-mean points) against the analytic shape  $g(J)$  (dashed), with a maximum deviation of 0.041 over the sweep. Panel (b) compares normalized shapes only: the absolute ideal purities differ (0.8100 for sinc against 0.9818 for Gaussian phase matching).

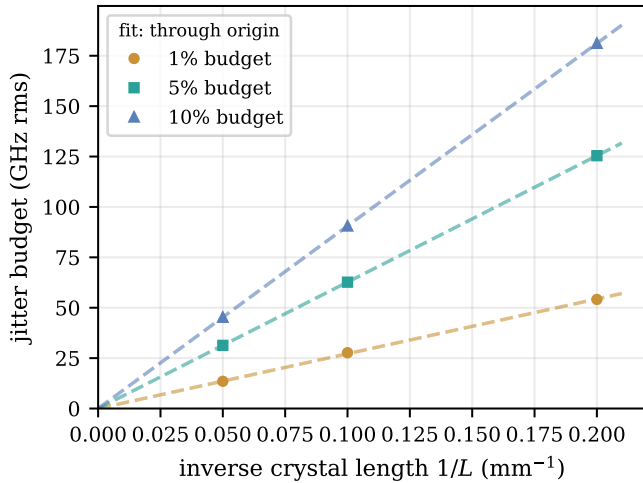


FIG. 5. Jitter budgets versus inverse crystal length for the three length-matched sources of Fig. 3. Dashed lines are linear fits through the origin for each relative-drop level; the 5% budgets stand in the ratios 1 : 2.00 : 4.00 at  $L = 20/10/5$  mm, demonstrating the  $1/L$  scaling of the jitter tolerance.

bandwidth in Eq. (2), and it gives the designer a one-parameter handle: halving the crystal length doubles every jitter budget.

### E. Longitudinal-mode structure

Multimode pump operation is far less forgiving. Figure 6(a) shows the purity ladder for an equal-weight comb at 0.25 THz spacing:  $P = 0.810, 0.671, 0.543, 0.445$ ,

0.377, 0.330, 0.292 for 1–7 lines. A single extra mode at this spacing already costs 17% of the purity—more than the entire 10% budget—so the maximum mode count within any budget is one. Figure 6(b) sweeps the spacing of a fixed 3-mode comb: at 0.02 THz the lines are unresolved by the JSA and the purity is essentially the single-mode value, while at 0.6 THz it has fallen to roughly 0.31. The spacing budgets are  $29.6 \pm 0.1$ ,  $75.9 \pm 0.5$ , and  $112.1 \pm 0.7$  GHz for 1/5/10%. The practical reading is that a nominally single-frequency pump that occasionally hops or lases on a side mode is a first-order threat to heralded purity, on par with the strongest jitter considered above.

### F. Intensity noise attacks a different metric

Relative intensity noise behaves qualitatively differently (Fig. 7). Because a pulse-energy fluctuation rescales the ensemble weights  $w_k$  in Eq. (4) without changing any normalized spectral amplitude, the heralded purity is exactly invariant: on the  $n = 64$  grid used for this study it is flat at 0.809956 for every RIN value from 0 to 30% rms (Fig. 7(a)). What RIN does damage is the pair statistics relevant for heralding fidelity and accidental double pairs [9, 14]: the double-pair boost  $\langle w^2 \rangle / \langle w \rangle^2$  follows the small- $r$  prediction  $1 + r^2$ , with a measured 1.0221 at 15% RIN against a theoretical 1.0225, and 1.0940 at 30% against 1.09 (Fig. 7(b)). The clipping of the weights  $w = \max[0, 1 + \mathcal{N}(0, r^2)]$  at zero shifts the expected boost by only  $\sim 10^{-4}$  even at 30% RIN (exact clipped value 1.0899), so the residual  $\pm 0.004$  scatter about  $1 + r^2$  is finite-sample noise of the  $N = 10^3$ , 8-seed estimator. Budgeted on the boost ( $N = 10^3$  realizations

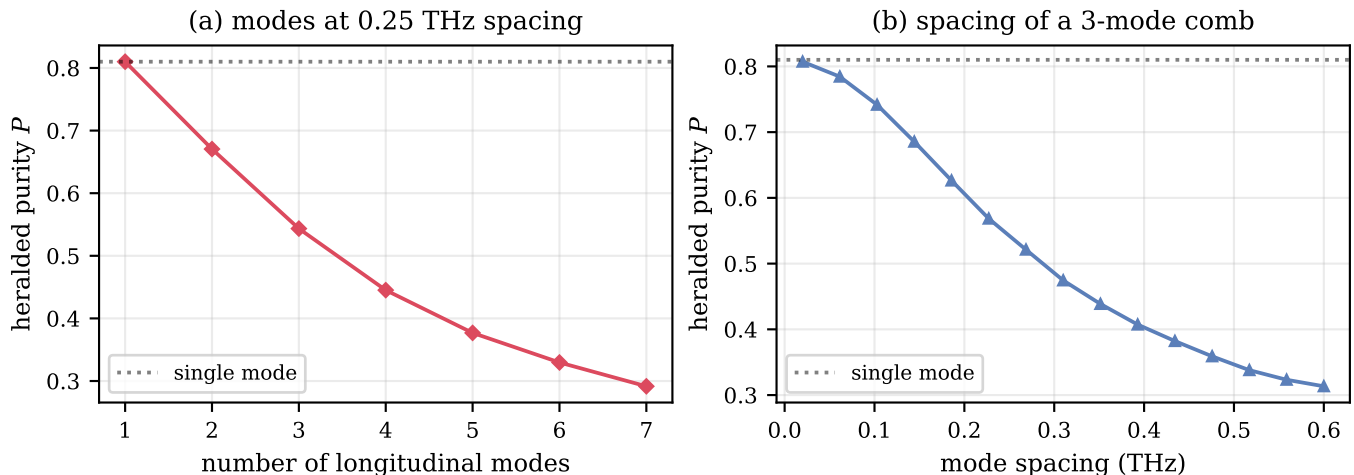


FIG. 6. Longitudinal-mode structure of the pump. (a) Heralded purity versus the number of comb lines at 0.25 THz spacing; the dotted line is the single-mode value. (b) Purity versus the spacing of a 3-mode comb (0.02–0.6 THz); for spacings unresolved by the JSA the single-mode purity is recovered.

per point, mean  $\pm$  SEM over 8 seeds), a +1% increase is reached at  $9.93 \pm 0.08\%$  RIN and +5% at  $22.3 \pm 0.2\%$ ; +10% is not reached within the 30% sweep. The practical cost is easiest to state through the coincidences-to-accidentals ratio (CAR). With mean pair number  $\mu$  per pulse, true heralded coincidences scale as  $\mu$  while accidental double pairs scale as  $\mu^2$ , so  $\text{CAR} \sim 1/\mu$  in the standard multi-pair framework [14]; RIN multiplies the double-pair term by the boost, dividing the CAR by it—equivalent to running the source at  $\mu_{\text{eff}} = \mu \times \text{boost}$ . At  $\mu = 0.05$ , a 30% RIN (boost 1.0940) costs 8.6% of the CAR, equivalent to  $\mu_{\text{eff}} = 0.0547$ ; at the +1% budget (9.9% RIN) the CAR cost is 1%.

#### G. The imperfection budget

Table II collects the results of Figs. 2–7 into a single auto-generated budget of eight rows. The convention is as follows: each entry is the largest value of the imperfection for which the heralded purity stays within a 1/5/10% *relative* drop of the ideal value of the same source (0.8100 for the 10 mm source, 0.8183 and 0.8061 for the re-matched 5 and 20 mm sources, and 0.9818 for the Gaussian phase-matching benchmark); the pump-bandwidth row is instead the window of bandwidths within the stated drop of the peak, and RIN—which leaves purity exactly invariant—is budgeted on the double-pair boost instead. The side-mode row counts the additional equal-power longitudinal modes at 0.25 THz spacing that fit each budget: zero additional side modes fit any budget, since a single extra mode already costs 17% of the purity. The hierarchy is unambiguous: longitudinal-mode structure is the least tolerable imperfection, center jitter comes next with budgets in the tens of GHz, and RIN only matters at the ten-percent level and through a dif-

ferent observable.

## IV. DISCUSSION

The central practical message of this study is that a pump-laser datasheet splits into two distinct damage channels, quantified row by row in Table II. Spectral specifications—center-frequency stability and longitudinal-mode structure—mix inequivalent Schmidt decompositions in the ensemble average of Eq. (4) and degrade the heralded purity. Statistical specifications—intensity noise—rescale each realization without changing its spectral shape, leaving the purity invariant while the double-pair boost inflates accidental coincidences through the multi-pair channel analyzed in Ref. [14]. Selecting a pump therefore means checking both columns: a spectrally quiet laser with high RIN delivers pure but noisy heralds, while an intensity-stable laser with a wandering center frequency delivers clean statistics on a mixed state.

The budgets also show where a designer has leverage. Jitter tolerance scales as  $1/L$ , a scaling we now demonstrate directly rather than infer: across crystals of 20, 10, and 5 mm (pump re-matched in each case) the 5% budgets of  $31.3 \pm 0.4$ ,  $62.7 \pm 0.7$ , and  $125 \pm 1$  GHz rms stand in the ratios 1 : 2.00 : 4.00, and through-origin linear fits in  $1/L$  describe all three drop levels within twice the seed error (Fig. 5; uncertainties throughout are mean  $\pm$  SEM over 16 Monte-Carlo seeds, 8 seeds for the RIN sweep). Halving the crystal from 10 mm to 5 mm (pump re-matched to 0.328 THz) thus doubles every budget, from  $27.7 \pm 0.3/62.7 \pm 0.7/91 \pm 1$  GHz rms to  $54.1 \pm 0.9/125 \pm 1/181 \pm 3$  GHz rms at the 1%/5%/10% levels (Fig. 3), at the price of a lower pair rate that our model does not capture. The ideal purity itself is un-



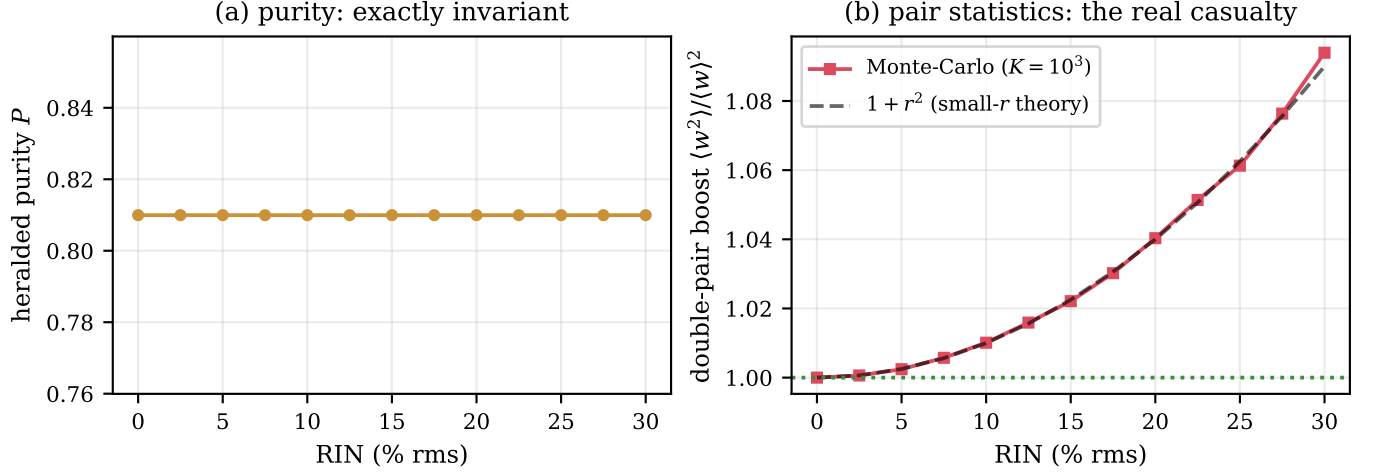


FIG. 7. Relative intensity noise. (a) Heralded purity versus RIN (0–30% rms): exactly invariant, flat at 0.809956 on the  $n = 64$  grid. (b) Double-pair boost  $\langle w^2 \rangle / \langle w \rangle^2$  from the Monte-Carlo ensemble ( $N = 10^3$  realizations per point) against the small- $r$  theory  $1 + r^2$  (dashed); the dotted line marks the noiseless value of 1.

| Imperfection                           | unit  | budget for relative purity drop |                |                 |
|----------------------------------------|-------|---------------------------------|----------------|-----------------|
|                                        |       | 1%                              | 5%             | 10%             |
| Center jitter rms (20 mm)              | GHz   | $13.5 \pm 0.2$                  | $31.3 \pm 0.4$ | $45.3 \pm 0.6$  |
| Center jitter rms (10 mm)              | GHz   | $27.7 \pm 0.3$                  | $62.7 \pm 0.7$ | $91 \pm 1$      |
| Center jitter rms (5 mm)               | GHz   | $54.1 \pm 0.9$                  | $125 \pm 1$    | $181 \pm 3$     |
| Center jitter rms (10 mm, Gaussian PM) | GHz   | $30 \pm 0.3$                    | $68.5 \pm 0.8$ | $102 \pm 1$     |
| Mode spacing (3-mode comb)             | GHz   | $29.6 \pm 0.1$                  | $75.9 \pm 0.5$ | $112.1 \pm 0.7$ |
| Additional side modes (0.25 THz)       | count | 0                               | 0              | 0               |
| Pump bandwidth window                  | THz   | 0.159–0.202                     | 0.131–0.244    | 0.114–0.281     |
| RIN (budget on pair boost)             | % rms | $9.93 \pm 0.08$                 | $22.3 \pm 0.2$ | beyond sweep    |

TABLE II. Imperfection budget, auto-generated from the sweeps of Sec. III. Each entry is the largest imperfection keeping the heralded purity within the stated relative drop of the ideal value of the same source. Monte-Carlo cells are quoted as the mean  $\pm$  SEM of the per-seed threshold crossings ( $S = 16$  seeds;  $S = 8$  for the RIN row); a cell is quoted only when every seed crosses inside the sweep, the confidence interval clears the zero-effect point, and the purity drop is resolved at  $2\sigma$ —otherwise a one-sided bound would be shown instead. Deterministic rows (pump bandwidth) carry no Monte-Carlo error. The bandwidth row gives the tolerance window around the purity peak, and RIN is budgeted on the double-pair boost  $\langle w^2 \rangle / \langle w \rangle^2$  because the purity is exactly RIN-invariant.

changed by this trade: the first-order JSA of Eq. (1) is invariant under the rescaling  $L \rightarrow L/2$ ,  $\sigma \rightarrow 2\sigma$  up to a common dilation of the frequency axes, which leaves the Schmidt spectrum—and hence the purity—exactly unchanged, and the slightly higher value quoted for the 5 mm sweep (0.818 versus 0.810) is a numerical windowing artifact—the fixed frequency window truncates relatively more of the sinc side lobes of the twice-wider 5 mm JSA—not a physical improvement. Bandwidth, by contrast, is the easy specification: the 1% window spans 0.159–0.202 THz, a full width of roughly 26% of the reference bandwidth, and the 10% window spans 0.114–0.281 THz. Center stability is the hard one: the 1% budget of  $27.7 \pm 0.3$  GHz rms corresponds, via Eq. (10) at  $\lambda = 775$  nm, to a wavelength stability of only 55.5 pm ( $\approx 0.06$  nm). A pump may exceed its nominal bandwidth by tens of percent with negligible cost—though

narrowing below the nominal value is penalized sooner, because the reference bandwidth sits near the lower edge of the tolerance window—but its center must hold to a sub-0.1-nm tolerance.

### Reading a datasheet

The budgets above are stated in the model’s units—GHz rms of center jitter, THz of amplitude bandwidth, percent rms of intensity noise—whereas manufacturers print nanometer spectral widths, temperature and current coefficients, side-mode suppression ratios, and integrated noise percentages. The translation layer of Sec. II, Eqs. (9)–(13), closes this gap, and Table III applies it line by line to three real products, one from each class

of pump laser a designer is likely to consider. As a worked anchor for the bandwidth line: the reference  $\sigma/2\pi = 0.164$  THz corresponds, via Eq. (9), to an intensity FWHM of 0.273 THz, i.e. 0.55 nm at 775 nm by Eq. (10).

The free-running Fabry–Perot diode (Thorlabs L785P090) is the cautionary case: its datasheet publishes *none* of the budget-relevant lines—no spectral width, no mode spacing, no temperature coefficient, no RIN, no side-mode suppression ratio. The printed 775–795 nm center wavelength is unit-to-unit selection, not drift. A Fabry–Perot diode is multimode by construction, and Sec. III shows that a single additional equal-power mode already costs 17% of the purity, more than the entire 10% budget, so the class fails before any number is checked. The deeper lesson is the absence itself: such a laser cannot even be budgeted from its datasheet, and is unusable for this application without full external spectral characterization.

The stabilized DFB diode (eagleyard EYP-DFB-0780) is the opposite extreme: every spectral-stability line it publishes passes with enormous margin. Its 1 MHz maximum linewidth enters the quasi-static model only as a 0.5 MHz bound, more than four orders of magnitude below the  $27.7 \pm 0.3$  GHz budget (a factor of  $\sim 2 \times 10^{-5}$ ). Its temperature coefficient,  $d\lambda/dT = 0.06$  nm/K, translates by Eq. (10) to 29.9 GHz/K, so the full 1% jitter budget—read as a uniform drift via Eq. (11)—equals holding the case temperature to  $\pm 1.6$  K (and, through  $d\lambda/dI = 0.003$  nm/mA, the drive current to  $\pm 32$  mA), ordinary servo performance. Its side-mode suppression of at least 30 dB is a side-line weight of  $10^{-3}$ , a negligible admixture. Yet this spectrally exemplary laser is the wrong tool for the present purpose: a CW pump with MHz linewidth sits five orders of magnitude below the 0.159–0.202 THz bandwidth window, and the purity collapses toward the strongly anti-correlated quasi-CW regime—already  $P = 0.04$  at the 5 GHz narrowband anchor of Sec. III, and falling roughly in proportion to bandwidth below that. The DFB class is the wrong choice for pure heralded photons, though precisely the right one for CW-pumped entanglement sources, where center stability is what matters.

The picosecond mode-locked oscillator (Spectra-Physics Tsunami) is the natural class, and its datasheet shows which line binds. At the short end of its pulse-width range, a 2 ps  $\text{sech}^2$  pulse (time–bandwidth product 0.315) has an intensity-spectrum FWHM of 0.158 THz, i.e.  $\sigma/2\pi = 0.095$  THz via Eq. (9) with the  $\text{sech}^2$  spectrum treated as an effective Gaussian, below the bandwidth window: at the 10 mm reference crystal this alone gives  $P = 0.66$ . Bandwidth is the binding line—and a benign one, because the purity is recovered by re-matching the crystal to  $L \approx 17$  mm rather than by modifying the laser. Every other line passes trivially: the specified 0.2% rms intensity noise gives a double-pair boost excess of only  $4 \times 10^{-6}$ , and the 80 MHz repetition-rate comb has teeth far too finely spaced to be resolved by

the JSA.

### Consistency with published measurements

The physical premises of the budgets are borne out by published experiments, beginning with the multimode channel. Kwon *et al.* [11] pumped a 3 mm type-I BBO crystal with a cw multimode 405 nm diode laser ( $\sim 24$  longitudinal modes within a 0.67 nm FWHM) and found that the pump coherence structure is completely transferred to the biphoton interference: the  $N = 2$  de Broglie (NOON) fringes show periodic wave-packet revivals at the pump mode-spacing period with  $\sim 98\%$  visibility. Their signal–idler Hong–Ou–Mandel dip, by contrast, has visibility above 98% with no revival—pair-exchange interference is insensitive to the multimode structure. This is exactly the premise of our comb channel: the pump modes imprint on the two-photon state along the sum-frequency coordinate, where they leave signal–idler interference untouched but directly damage the quantity our budgets bound, the heralded single-photon purity probed by *two-source* interference. Since Ref. [11] does not measure heralded purity, the comparison is premise-level rather than number-level.

For the observable itself, the flagship measurement is that of Mosley *et al.* [6]: two independent group-velocity-matched KDP sources, pumped by femtosecond pulses at 415 nm and used without narrowband spectral filters, gave a four-fold Hong–Ou–Mandel visibility of  $94.4 \pm 1.6\%$  between independently heralded photons—a raw value with no background subtraction, presented as a lower bound on the mean photon purity, at heralding efficiencies up to 44%. This is precisely the quantity our budgets constrain, since  $V_{\text{HOM}} = P$  for independent identical sources. A singular-value decomposition of the measured joint spectral intensity, under an assumed flat phase, instead gives  $\sim 98\%$ : even in a flagship source, real-world pump and phase imperfections cost several percent—exactly the 1–5% scale that the budget rows of Table II translate into datasheet tolerances.

Zielnicki *et al.* [10] make the same point from the characterization side: for one and the same unfiltered BBO source, the inferred purity is technique-dependent,  $0.88 \pm 0.02$  (Fourier-transform JSI) and  $0.87 \pm 0.03$  (dispersive-fiber JSI) against  $0.66 \pm 0.02$  (from  $g^{(2)}$ ) and  $0.61 \pm 0.05$  (two-source HOM). JSI-based methods are blind to the joint spectral phase—pump chirp in particular—and therefore overestimate the purity, which empirically motivates the stated scope of our model: a flat spectral phase throughout, with a chirp channel as a natural future extension [12]. We note honestly what these comparisons do not provide: none of these experiments publishes purity as a function of a datasheet specification—precisely the gap the budget tables fill—so the published record validates the premises and scales of the model, not the budget numbers themselves.

| Datasheet line (as printed)                                                                                                                                                                                                                       | Value                | Model translation                   | vs. budget                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------|-----------------------------------------|
| <i>Free-running FP diode</i> — Thorlabs L785P090 (785 nm; Thorlabs spec sheet 21159-S01 Rev. E (2022))                                                                                                                                            |                      |                                     |                                         |
| Center wavelength (unit-to-unit)                                                                                                                                                                                                                  | 775–795 nm           | 10 THz window                       | selection, not drift                    |
| Spectral width, Mode spacing / cavity length, $d\lambda/dT$ , RIN, SMSR                                                                                                                                                                           | not published        | —                                   | —                                       |
| <b>Fails by construction:</b> multimode with none of the budget-relevant lines published — unusable without external characterization.                                                                                                            |                      |                                     |                                         |
| <i>Stabilized DFB diode</i> — eagleyard Photonics EYP-DFB-0780 (780.24 nm; eagleyard datasheet Rev. 1.03 (2022))                                                                                                                                  |                      |                                     |                                         |
| Linewidth (FWHM)                                                                                                                                                                                                                                  | typ. 0.6, max. 1 MHz | $\leq 0.5$ MHz bound                | $2 \times 10^{-5} \times$ the 1% budget |
| Temp. coefficient $d\lambda/dT$                                                                                                                                                                                                                   | typ. 0.06 nm/K       | 29.9 GHz/K                          | 1% budget = $\pm 1.6$ K                 |
| Current coefficient $d\lambda/dI$                                                                                                                                                                                                                 | typ. 0.003 nm/mA     | 1.50 GHz/mA                         | 1% budget = $\pm 32$ mA                 |
| SMSR                                                                                                                                                                                                                                              | min. 30 dB           | side-line weight $1 \times 10^{-3}$ | negligible admixture                    |
| RIN                                                                                                                                                                                                                                               | not published        | —                                   | —                                       |
| <b>Spectral columns exemplary</b> (drift budget met by holding $\pm 1.6$ K), but CW linewidth $\sim$ MHz lies five orders below the bandwidth window: $P \rightarrow 0.04$ (CW limit) — wrong class for pure heralds, ideal for center stability. |                      |                                     |                                         |
| <i>ps mode-locked oscillator</i> — Spectra-Physics Tsunami (ps) (775 nm; Tsunami datasheet (MKS/Newport))                                                                                                                                         |                      |                                     |                                         |
| Pulse width (sech <sup>2</sup> )                                                                                                                                                                                                                  | <2–100 ps            | $\sigma = 0.095$ THz                | $P = 0.66$ at 10 mm                     |
| Repetition rate                                                                                                                                                                                                                                   | 80 MHz               | comb spacing 80 MHz                 | unresolved by the JSA                   |
| Noise (10 Hz–2 MHz)                                                                                                                                                                                                                               | <0.2% rms            | $r = 0.2\%$                         | boost $-1 = 4 \times 10^{-6}$           |
| Center wavelength stability                                                                                                                                                                                                                       | not published        | —                                   | —                                       |
| <b>The natural class:</b> drift, comb and RIN lines pass trivially; the binding line is bandwidth — 2 ps gives $\sigma = 0.095$ THz and $P = 0.66$ at 10 mm, recovered by re-matching the crystal to $L \approx 17$ mm.                           |                      |                                     |                                         |

TABLE III. Reading three real pump-laser datasheets against the imperfection budgets. Each printed specification (first two columns) is converted into the model’s units (third column) with the translation layer of Sec. II, Eqs. (9)–(13), and compared against the Monte-Carlo budgets of Table II (last column). Lines the manufacturers do not publish are marked as such—for the free-running diode the omissions are themselves the finding. Sources are the public datasheets cited in each block header.

### Extensions and limitations

Our ideal purity of 0.810 is the familiar ceiling of sinc-shaped phase matching, set by the side lobes of the unapodized crystal. Domain engineering and apodization reshape the phase-matching function toward a Gaussian and raise this ceiling well above 0.81 [7, 8]. Our budgets are *relative* degradations from whatever ceiling the phase matching provides, and the Monte-Carlo ensemble method applies unchanged to engineered crystals: only the sinc factor in Eq. (1) need be replaced. This is now demonstrated rather than asserted: repeating the full jitter sweep with the amplitude-matched Gaussian phase matching of Eq. (6), whose ideal purity is 0.9818, yields relative budgets of  $30 \pm 0.3/68.5 \pm 0.8/102 \pm 1$  GHz rms (Table II), about 8% *larger* than the sinc values at the 1% level (8–13% across the three levels), and these are reproduced exactly, within one standard error, by the closed form of Eq. (8), which predicts 29.9/69.0/101.7 GHz (Fig. 4). The sinc-model budgets quoted above are therefore mildly conservative for engineered crystals.

The budgets also translate into two-photon interference, but the translation depends on the architecture.

For two *independent* identical heralded sources the Hong–Ou–Mandel visibility equals the single-source purity,  $V_{\text{HOM}} = P$  [1, 6], so the jitter budgets above bound the interference of sources pumped by separate free-running lasers. If instead both sources are pumped by the *same* laser, the drift is common-mode: in each trial the two JSAs are translated rigidly together by the same center shift, every realization retains the ideal single-source purity, and within this model the shared-drift HOM visibility stays at the ideal value  $P_0$  *exactly*, independent of the jitter amplitude. The reported budgets are therefore conservative for single-laser multiplexed architectures, in which center jitter cancels in common mode.

Several limitations bound the present results, each pointing to future work. The model uses first-order GVM only:  $\Delta k$  is linearized, with no Sellmeier dispersion or higher-order terms, so the phase-matching ridge is perfectly straight; curvature will matter for broadband pumps and long crystals. We compute heralded purity only; heralding efficiency and absolute pair rate are outside the model, though a complete source assessment must weigh all three jointly [9]. The quasi-static treatment of jitter is no longer an open caveat: its regime

of validity is bounded explicitly in Sec. II—noise slower than the  $\sim 1$  ps pump coherence time and faster than the integration time—and only sub-picosecond noise, which acts coherently within each pulse, falls outside it on the fast side; drift slower than the integration time is instead summarized deterministically through the bounded-excursion conversion of Eq. (11). An incoherent pump linewidth, beyond the Gaussian coherent bandwidth, is not yet a separate mixed-state channel, which is why the Lorentzian linewidth in Table III enters only as a quasi-static bound. Similarly, the comb model asserts mutually incoherent lines and therefore covers free-running multimode diodes only; a mode-locked pump comb is phase-coherent, so its JSA is a coherent sum—a pure state—and requires a coherent-sum treatment rather than the mixture of Eq. (4). Finally, the treatment is collinear and single-spatial-mode.

## V. CONCLUSION

We have converted pump-laser datasheet specifications into a quantitative imperfection budget for the heralded spectral purity of an SPDC source, using a first-order GVM model with Monte-Carlo pump ensembles vali-

dated against a closed-form double-Gaussian benchmark that reproduces the measured Gaussian-phase-matching budgets exactly:  $30 \pm 0.3/68.5 \pm 0.8/102 \pm 1$  GHz measured versus  $29.9/69.0/101.7$  GHz predicted. For the reference source (10 mm crystal, ideal purity 0.810), keeping the purity within 1%/5%/10% of ideal requires center-frequency jitter below  $27.7 \pm 0.3/62.7 \pm 0.7/91 \pm 1$  GHz rms, and the budgets obey a demonstrated  $1/L$  law over 5–20 mm crystals (ratios 1 : 2.00 : 4.00), so halving the crystal doubles every budget at the cost of pair rate. Single-longitudinal-mode operation is mandatory: a single additional equal-power mode already costs 17% of the purity. Intensity noise, by contrast, leaves the purity exactly invariant and instead inflates the double-pair boost. The rule for choosing a pump is simple: spectral specifications damage purity, statistical specifications damage pair statistics, and both columns of the datasheet must be checked.

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